

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION**Annual Examination 2026
CHEMISTRY — SSC-I (9th Grade)****Total Marks: 65****Time Allowed: 2 Hours 30 Minutes****Version: 1****General Instructions:**

1. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
2. **Section B** (Short Questions): Attempt **11**, choosing main **OR** alternative. Each **3 marks**.
3. **Section C** (Long Questions): Attempt **4**, choosing main **OR** alternative. Each **5 marks**.
4. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)*Encircle the correct option. All questions are compulsory.*

#	Question	A	B	C	D	Answer
1	Chemistry deals with the composition, structure, properties, and changes of:	Energy	Space	Matter	Light	(A) (B) (C) (D)
2	The branch that deals with carbon-containing substances (except carbonates, bicarbonates, oxides, carbides) is:	Inorganic	Physical	Organic	Analytical	(A) (B) (C) (D)
3	The branch that focuses on polymers, their types, properties, and uses is:	Industrial	Polymer	Nuclear	Astro	(A) (B) (C) (D)
4	The integration of scientific knowledge for human needs is known as:	Science	Engineering	Technology	Research	(A) (B) (C) (D)
5	Forensic chemistry is the application of:	Organic	Analytical	Physical	Environmental	(A) (B) (C) (D)
6	Anything that has mass and occupies space is called:	Energy	Element	Compound	Matter	(A) (B) (C) (D)
7	A highly ionized gas where electrons are separated from atoms is:	Solid	Liquid	BEC	Plasma	(A) (B) (C) (D)
8	Bose-Einstein Condensate (BEC) occurs at temperatures near:	100°C	0°C	-273°C	25°C	(A) (B) (C) (D)
9	Diamond, graphite, and buckyballs are allotropes of:	Silicon	Oxygen	Carbon	Nitrogen	(A) (B) (C) (D)
10	A solution containing maximum solute at a given temperature is:	Unsaturated	Supersaturated	Dilute	Saturated	(A) (B) (C) (D)
11	The scattering of light by colloidal particles is called:	Reflection	Refraction	Tyndall effect	Dispersion	(A) (B) (C) (D)
12	Water is called a universal solvent because it:	Is colorless	Dissolves most compounds	Is everywhere	Has no taste	(A) (B) (C) (D)

SECTION B — Short Questions (11 × 3 = 33 Marks)*Attempt 11 questions. Choose either the main question OR its alternative.*

Q.	Question	Marks		OR Question	Marks
2(i)	Define chemistry. List any four branches of chemistry with their definitions.	3	OR	What is green chemistry? Why is it important in reducing pollution?	3
2(ii)	Differentiate between organic chemistry and inorganic chemistry.	3	OR	Differentiate between geochemistry and astrochemistry.	3
2(iii)	Differentiate between science and technology with examples.	3	OR	Define engineering. How do chemical engineers contribute to industry?	3
2(iv)	Explain with an example how science, technology, and engineering work together.	3	OR	Describe two daily life applications of chemistry from different branches.	3
2(v)	Define matter. List four states of matter with one example each.	3	OR	What is BEC? What are liquid crystals? Give one example of each.	3
2(vi)	Compare macroscopic properties (density, compressibility, fluidity) of solids, liquids, and gases.	3	OR	Define plasma. Give three examples of plasma in nature or daily life.	3
2(vii)	Define element, compound, and mixture with one example each.	3	OR	Differentiate between compound and mixture (any three differences).	3
2(viii)	What is allotropy? Name and briefly describe three allotropes of carbon.	3	OR	Describe the structure and properties of diamond. Why is it non-conductive?	3
2(ix)	Define solution, solute, and solvent. Give one example of each.	3	OR	What is an aqueous solution? Why is water called a universal solvent?	3
2(x)	Differentiate between saturated and unsaturated solutions.	3	OR	What is a supersaturated solution? How can it be prepared?	3
2(xi)	Differentiate between colloid and suspension (any three differences).	3	OR	Explain the effect of temperature on solubility of solutes with examples.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)*Attempt 4 questions. Choose either the main question OR its alternative.*

Q.	Question	Marks		OR Question	Marks
3(i)	Describe any five branches of chemistry with their definitions and one daily life application of each.	5	OR	Explain the relationship between science, technology, and engineering with two detailed examples from chemistry.	5
3(ii)	Describe the states of matter in detail including solid, liquid, gas, and plasma with properties and examples.	5	OR	What are allotropes? Explain diamond, graphite, and buckyballs in detail with structure and properties.	5
3(iii)	Differentiate between elements, compounds, and mixtures with examples. Draw the classification of matter flowchart.	5	OR	What is a solution? Describe gaseous, liquid, and solid solutions with at least two examples each.	5
3(iv)	Explain saturated, unsaturated, and supersaturated solutions. Discuss the effect of temperature on solubility.	5	OR	Compare solutions, colloids, and suspensions including particle size, visibility, filtration, Tyndall effect, and stability.	5

Invigilator's
SignatureCandidate's
Signature**Chapters Covered:** Ch 1 — Nature of Chemistry in Science | Ch 2 — Matter**— End of Question Paper —**